

Critical AI Information Literacy

Foundations of AI Workshop Series · Workshop 4

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Winter 2026

Today You'll Learn

- Why GenAI outputs need evaluation
- How to identify common limitations
- Practical verification strategies
- Responsible use in academic settings

Your Journey So Far

W1



NLP to LLMs

Tokenization, embeddings,
attention, next-token prediction

W2



Deep Learning

Neural networks, CNNs,
LSTMs, diffusion models

W3



GenAI Brainstorming

Model selection, prompting,
parameters, DartmouthChat

Today: You know how GenAI works and how to use it. Now we learn how to evaluate what it produces.

Today's Journey

Content

- Common GenAI limitations
- The CHECK evaluation framework
- Understanding bias in outputs
- Academic use and responsible practices

4 Dartmouth Chat Activities

The Citation Hunt

Image Generation

Multi-Model Face-Off

Bias & Perspectives

Common Limitations of GenAI



Hallucinations

Generating plausible but incorrect information, including fabricated citations and statistics. Improving with newer models, but still occurs.



Knowledge Boundaries

Training data cutoffs, gaps in niche domains, and limited awareness of recent events. Models don't always signal uncertainty.



Bias Reflection

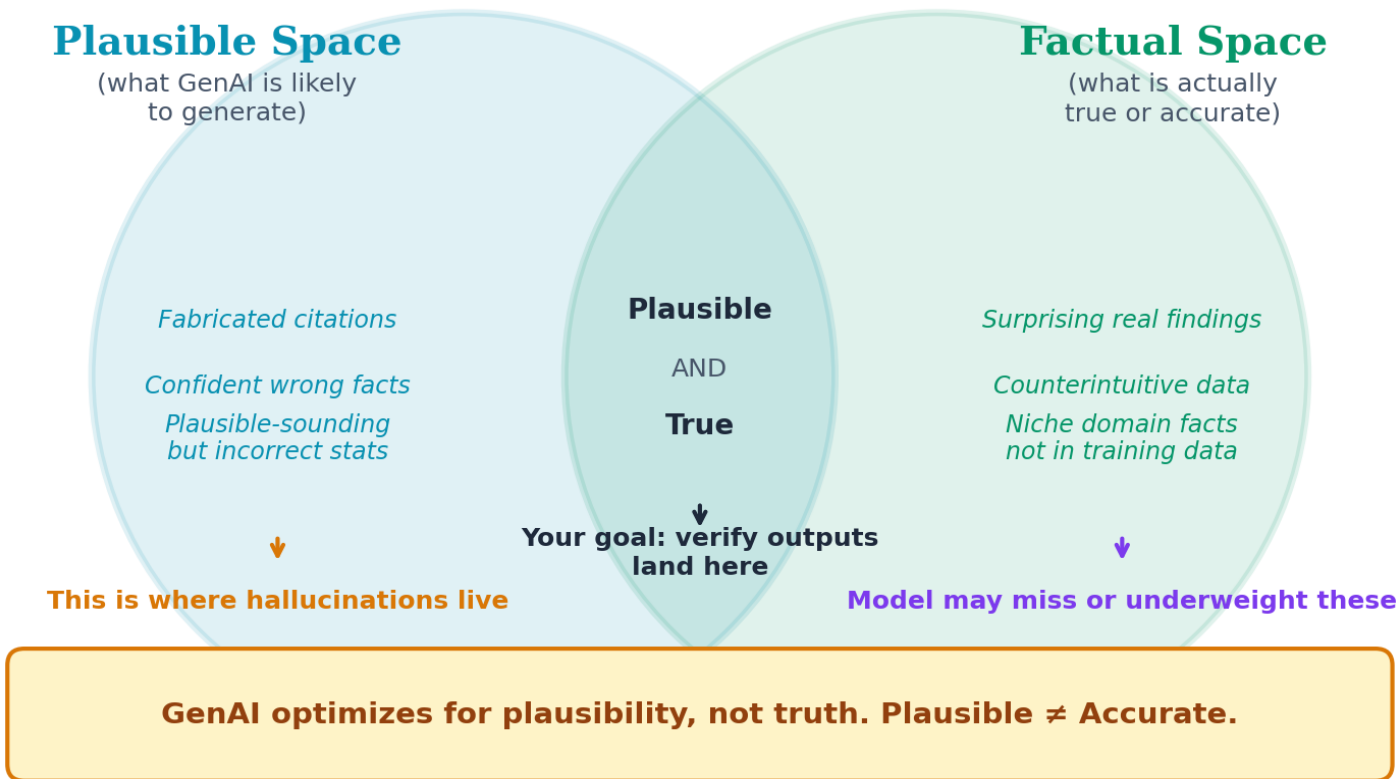
Training data contains societal patterns. Models can reproduce stereotypes related to gender, race, culture, often subtly.



Reasoning Errors

Mistakes in logic, math, or multi-step problems. Models may take shortcuts that appear correct but aren't.

Plausible ≠ Accurate



Activity 1: The Citation Hunt (10 mins)

ACTIVITY

1. Pick a Topic

Choose something in your area of expertise.

2. Ask for Citations

Go to chat.dartmouth.edu and pick a model

Prompt: "Write a paragraph about [topic] with 3-4 academic citations including authors, year, and journal."

3. Verify

Search each reference in Google Scholar. Do the authors exist? Did they write that paper? Is the journal real?

4. Compare Models

Try the same prompt on a different model. Does one fabricate more than the other?

The CHECK Framework

A practical checklist for evaluating any GenAI output

Claims

What specific assertions does the output make?

How Generated

This was produced probabilistically. Is that sufficient here?

Evidence

Can you trace claims to verifiable, authoritative sources?

Consistency

Does it contradict itself or conflict with known information?

Knowledge Gaps

What might be missing from the model's training data?

Activity 2: Image Generation (10 mins)

ACTIVITY

1. Generate

Use an image model. Try a prompt requiring accuracy: "A human hand holding a coffee cup" or "A map of New England with labels"

2. Evaluate

Anatomical errors? Incorrect text? Extra fingers? Missing details? How faithful is it to your prompt?

3. Test Bias

Try "a scientist" or "a nurse" or "a CEO" without specifying demographics. What does the model assume?

4. Reflect

Where would you trust this output? Where would you need to verify? How does this compare to text limitations?

Bias, Transparency, and Responsible Use



Understanding Bias

Training data reflects societal patterns.

Models learn and can amplify these patterns.

Outputs may contain subtle biases.

What you can do:

- Test with diverse perspectives
- Question model defaults and assumptions
- Use system prompts to broaden output
- Consider whose views are represented



Responsible Use

Transparency

Disclose when and how GenAI was used

Accountability

Check model responses rigorously

Context

Policies vary across departments and journals

Activity 3: Multi-Model Face-Off

ACTIVITY

Compare a smaller vs larger model on the same prompt. Use 2 models at a time.

1. Pick a Prompt

Try one of these or use your own:

"Explain why antibiotics don't work on viruses"

"A patient has fever of 103 and a rash. What are 3 possible diagnoses?"

"Write a proof that there are infinitely many prime numbers"

"Summarize the key differences between GDP and GNP with examples"

2. Run Side by Side

Pick a smaller and larger model (e.g., Haiku vs Opus, GPT-5-instant vs thinking, Gemini Flash vs Pro). Submit the same prompt to both.

3. Compare

Does the larger model provide better reasoning or fewer errors? Is the smaller model good enough for the task? Where do they disagree?

Activity 4: Bias & Perspectives

ACTIVITY

1. Test Defaults

Prompt: "Write a short story about a doctor" or "Describe a typical CEO." What does the model assume?

2. Reframe

Add specifics: "a doctor in rural Kenya" or "a 25-year-old CEO from Brazil." How does the output shift?

3. Compare Models

Ask a culturally nuanced topic to two models. Do they handle nuance differently?

4. System Prompt

Set system prompt: "Consider diverse global perspectives." Re-run your prompts. Does the output improve? Share findings with the group.

Key Takeaways



GenAI is powerful and improving. Critical evaluation makes you a better user.



Use CHECK: Claims, How generated, Evidence, Consistency, Knowledge gaps.



Always verify citations, statistics, and domain-specific facts independently.



Compare models, ask for self-critique, and break complex tasks into pieces.



Be transparent about GenAI use and follow your institution's guidelines.

Thank You!

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Resources

Dartmouth Chat: chat.dartmouth.edu

Follow-up Questions → mashrekur.rahman@dartmouth.edu

Series Recap

W1: How GenAI processes language

W2: Deep learning and computer vision

W3: Model selection and prompting

W4: Critical evaluation and literacy ✓